

Component C — The Statistical Layer

Group 5

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1. The Three Metric Definitions

1. Winner Correct Rate

It is a performance metric that measures how accurately an ML model predicts the winning outcome among other possible classes.

Formula:

$$\text{WCR} = \text{Number of replicates in which the declared winner is truly the best} / \text{Total number of valid replicates}$$

For instance, consider a model designed to predict the winner of the derby horse race. Suppose the model makes predictions for 20 matches. If the model predicts the winning horse accurately for 15 times, and for the remaining 5 times the model fails to predict the winning horse accurately. then the WSR turns out to be as:

$$\text{WCR} = 75/100 = 75\%$$

Therefore, the model has an accuracy of 75%.

A higher WSR means the model is more successful in identifying the correct outcome and vice versa.

2. Kendall's Tau (tau):

It is a statistical measure that evaluates the magnitude and direction of the relationship between two ranked variables. It measures the degree at which two ranked variables are related to each other.

Kendall's Tau is useful for ranked and ordinal data. It does not require a linear relationship between variables.

The coefficient is calculated based on concordant pairs and discordant pairs:

$$\tau = (C - D) / nC2$$

where, C is number of concordant pairs, D is number of discordant pairs, n is total number of observations.

When the relative ranking of two observations is the same for both variables it is called a concordant pair.

For instance, 5 horses participate in the derby race. A model predicts the ranking before the match. After the match the actual rankings are as follows:

Horse	ML Predicted Rank	Actual Rank
A	1	1
B	2	2
C	3	4
D	4	3
E	5	5

Here, the model correctly identified the ranking for most of the horses, but the prediction for horse C and D is interchanged.

For this race, the total number of possible ranking pairs are $n(n-1)/2 = (5*4)/2=10$ pairs.

After comparison there are 9 concordant and 1 discordant pair.

$$\tau = (C-D)/10$$

$$\tau = (9-1)/10$$

$$\tau = 0.8$$

The score of 0.8 shows a strong positive correlation between the predicted rankings and the actual rankings.

A Kendall's Tau value of $\tau=1$ is a perfect match between predicted ranking and actual ranking, $\tau=0$ indicates no correlation between the rankings, $\tau=-1$ indicates negative correlation.

Therefore, Kendall's Tau can be used to evaluate how well the model preserves the overall ordering of outcome, rather than considering only whether it correctly predicts the winner.

3. Optimism Bias

It is the difference between the reported performance score by the value and the true performance.

Optimism Bias=Reported Winning Score–True Score of the Selected Winner

Note: The true score being subtracted is the true performance of the model selected as the winner, not the true score of the genuinely best model.

For instance, three models are used to predict the rankings of the Derby horse race.

Model	Reported Winning Score	True Score
Model A	82%	76%
Model B	79%	78%
Model C	75%	74%

Here, model A is selected as the winning model, because it has the highest reported score of 82%. Its true score, evaluated on independent or unseen data, is 76%.

Optimism Bias = 82% - 76% = 6%

This overestimates the performance of the model by 6%.

Thus, Optimism Bias specifically measures the gap between the reported performance of the selected winner and the true performance of that same selected winner. It does not measure how far the selected model is from the genuinely best-performing model.

4. Power for the c1 vs c2 Comparison

Power measures the proportion of valid replicates in which the statistical test correctly detects that classifier c1 performs better than classifier c2.

For each replicate, the five fold-level AUCs of c1 are paired with the corresponding five fold-level AUCs of c2. A paired t-test is then performed using the paired fold results.

A correct detection occurs when:

- $p < 0.05$, and
- The mean paired difference favors c1.

A sign error occurs when:

- $p < 0.05$, but
- The mean paired difference favors c2.

Power is calculated as:

Power = Number of correct detections / Total number of valid replicates

The sign-error proportion is calculated as:

Sign Error = Number of sign errors / Total number of valid replicates

If the paired t-test cannot be performed, the replicate produces no detection and contributes 0 to both power and sign-error calculations.

2. The Edge-Case Catalog

Edge case	Pre-registered rule	Justification
Two classifiers tie for 1st place in a study replicate	Choose the lexicographically smallest <code>classifier_id</code> as the declared winner.	This provides a deterministic rule and avoids arbitrary or random selection between tied classifiers.
A single-class fold yields an undefined AUC	Treat the fold as invalid and do not replace the AUC with 0. If a classifier has fewer than 5 valid folds, the replicate is invalid.	AUC cannot be computed meaningfully for a fold containing only one class. Replacing it with 0 would introduce an artificial performance value.
An incomplete replicate or invalid fold structure	A valid replicate must contain exactly 5 valid, unique folds with fold IDs $\{1,2,3,4,5\} \setminus \{1,2,3,4,5\}$ for every required classifier. Duplicate, missing, out-of-range, or non-integer fold IDs invalidate the replication.	This ensures that every classifier is evaluated using the same complete five-fold structure.
Missing true AUC in the Truth Arm	The truth file must contain a valid true AUC for every classifier in the replicate, matched using the exact <code>dataset_id</code> and <code>classifier_id</code> . Otherwise, the replication is invalid.	A true ranking cannot be determined without a valid ground-truth AUC for every classifier being compared.

Missing, non-numeric, or NaN AUC	Treat the corresponding fold as invalid. If fewer than 5 valid unique folds remain for the classifier, exclude the replicate.	Invalid numerical values cannot be used to calculate a reliable replicate-level mean.
Paired t-test cannot be performed	No detection is recorded; power = 0 and sign-error = 0 for that replicate.	A replicate cannot provide evidence for either a correct detection or a sign error when the test cannot be performed.
$p=0.05$ $p=0.05$ exactly	Record no detection because the detection criterion is strictly $p<0.05$.	This follows the pre-specified significance threshold without treating equality as significant.
Mean paired difference = 0	Record no detection and no sign error.	If the observed paired difference is exactly zero, neither classifier is favored by the direction of the comparison.

3. Code Verification Plan

Before running the analysis on the synthetic experimental results, we'll test the calculation scripts using a small synthetic file where the correct result is already known. This helps us catch errors in the ranking, bias, and statistical test calculations.

1. Planned Toy Verification

We'll create a small toy dataset containing three classifiers with known true AUC values and five fold-level AUC values for one replicate.

The true AUC values are:

Classifier	True AUC
c1	0.85
c2	0.82
c3	0.76

For the study to replicate, the five fold AUCs are averaged to obtain the observed mean AUC for each classifier.

The expected results are:

- c1 mean AUC = 0.810
- c2 mean AUC = 0.830
- c3 mean AUC = 0.754

Therefore, the observed ranking is:

$c2 > c1 > c3$

while the true ranking is:

$c1 > c2 > c3$

The expected outputs from the script are:

Statistic	Expected Result
Winner-correct rate	0
Kendall's τ	0.333333
Optimism	0.010000
Power	0
Sign-error proportion	0

The script should reproduce these manually calculated values.

2. Hand Calculation

For classifier c1:

$$0.81+0.78+0.84+0.80+0.825=0.810$$

For classifier c2:

$$0.86+0.83+0.80+0.85+0.815=0.830$$

For classifier c3:

$$0.74+0.77+0.72+0.79+0.755=0.754$$

The observed winner is therefore c2, while the true best classifier is c1.

Hence:

$$\text{Winner-correct}=0$$

The observed ranking is $c2 > c1 > c3$, while the true ranking is $c1 > c2 > c3$, giving:

$$\tau=0.333333$$

For optimism, the true AUC of the declared winner c2 is used:

$$\text{Optimism}=0.830-0.820=0.010$$

For the c1 versus c2 comparison, the paired fold differences are:

$$[-0.05, -0.05, +0.04, -0.05, +0.01]$$

The mean difference is $-0.02-0.02$, and the paired t-test does not produce a significant detection at $p<0.05$. Therefore:

$$\text{Power}=0$$

and

Sign-error=0

3. Automated Verification

The calculation script is also tested using automated unit tests covering the metric calculations, data validation, edge cases, and evaluation at the specified sample sizes.

Part 2: The Ambiguity List

1. Statistical Test for Power: The instructions ask us to calculate statistical power but do not name the exact test to use. We assume we should use a standard two-sided paired t-test for most protocols, and only use the Nadeau-Bengio test for Protocol 6.
2. Significance Boundary: The rules mention finding a "significant" difference but leave out the exact p-value cutoff. We assume the cutoff is strictly $p < 0.05$, meaning exactly $p = 0.05$ counts as "No Detection."
3. Defining the "Gap": It is unclear which two models we use to calculate the "true gap" for statistical power. We assume this gap is always the difference between the true best and true second-best models based on the large Truth Arm dataset.
4. Aggregating Repeated CV: Protocol 3 generates 50 different AUC scores for each classifier, but the instructions do not say how to combine them. We assume we should take the simple average (mean) of all 50 scores to pick the overall winner.
5. Optimism Bias with Ties: If two models tie for the highest score, the instructions do not explain which one's true score to use for the Optimism Bias calculation. We assume we should just pick the model with the alphabetically smaller ID to break the tie.
6. AUC Calculation Method: The project asks us to use "AUC" but does not specify if it means standard ROC AUC or Precision-Recall AUC (which matters for imbalanced data). We assume we must use the standard ROC AUC for every dataset.

7. Handling Incomplete Replicates: The rules do not explain what to do if a fold fails and returns an undefined or missing AUC score. We assume that if any fold fails, that model's run is invalid, and the entire replicate must be thrown out.
8. Power Calculations on Zero Variance: The instructions do not say how to record a power test if the math fails or if the difference between models is exactly zero. We assume these broken or zero-difference tests automatically count as "No Detection."